Technical Implementation Plan

Canton Application Layer | Prediction Markets

TruthOracle: Technical Implementation Plan

A scoped engineering plan implementing the TruthOracle protocol: dynamic pm-AMM liquidity, tiered objective resolution, and the prediction-model marketplace, delivered across four months with defined milestones and acceptance criteria.

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Prepared for: Cantor8
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Implements: TruthOracle Proposal v0.7
Target SDK: Daml 3.4.x / Canton

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Window: Weeks 1–16 (4 months from commencement)

1 Purpose and Scope

This document is the technical architecture and implementation plan for delivering the TruthOracle prediction market protocol as specified in the proposal it implements. It is written to be attached as an annex to a delivery agreement: it enumerates the work as discrete **workstreams**, schedules them across a **four-month** (sixteen-week) window, and binds each **milestone** to objective **acceptance criteria**. Where a behaviour is specified in the proposal, this plan references it rather than restating it.

In scope. Implementation of the four core Daml contracts and the optional prediction-model marketplace; the tiered resolution mechanism (T0–T3) and its off-ledger resolver/reporter services; the trading frontend and Cantor8 Enterprise Wallet integration; continuous integration, DAR build and versioning; a Canton DevNet pilot; compliance and observability tooling; a security review; and a MainNet-readiness package.

Out of scope. Listed explicitly in §10. In summary: regulatory licensing, the production MainNet launch itself (gated on that licensing), KYC-provider build-out, fiat on/off ramps, native mobile applications, and collateral-token issuance.

The protocol surface, contract semantics, pricing model, resolution tiers, and risk model are defined in the proposal and are not re-derived here.

2 Architecture Overview

The system is a layered application on Canton. All trust-bearing state and logic live in the Daml contract layer; off-ledger services orchestrate fetching, signing, and presentation but hold no authority the ledger does not grant them.

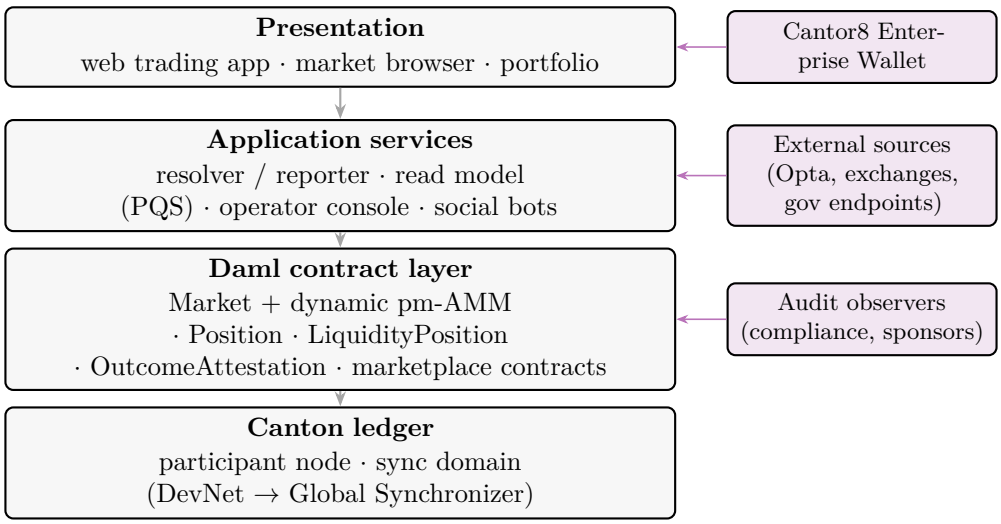


Figure 1: Layered architecture. Trust resides in the Daml contract layer; the services and presentation around it are replaceable.

Component inventory.

Component	Responsibility
Daml core package	The `Market` (with inline *dynamic* pm-AMM reserves and the Gaussian pricing math — normal-distribution $\Phi/\phi/\Phi^{-1}$, fixed-point helpers, and the invariant / optimal-reserve solver), `Position`, `LiquidityPosition`, and `OutcomeAttestation` templates, the `ResolutionPolicy` record, and shared types.
Resolution logic	The `Resolve`, `ProposeOutcome`, `Challenge`, `Finalize`, and `Void` choices implementing tiers T0–T3, including on-ledger `secp256k1` verification for T1.
Resolver / reporter service	Off-ledger service that fetches an allowlisted source over TLS, parses the outcome, signs, and creates an `OutcomeAttestation`; one instance per allowlisted reporter.
Read model (PQS)	Off-ledger indexer over the Participant Query Store that projects market state — price history, volume, open interest, trade and resolution metrics, and per-market position books — for the trading UI, dashboards, and audit observers; holds no ledger authority.
Operator console & key management	Operator tooling for market creation, `ResolutionPolicy` authoring, and audit-observer administration (operator and treasury jointly), plus custody and rotation of source (T1) and reporter (T2) signing keys.
Marketplace package	`PredictionModel`, `ModelForecast`, `ModelPrizePool`, scoring/distribution, and the "The Model" aggregation.
Frontend & wallet	Trading UI and Cantor8 Enterprise Wallet integration, including the two-step collateral-transfer/reconciliation flow.
Compliance & observability	Audit-observer projection, position-book reporting, and operational dashboards.

3 Workstreams

Delivery is organised into six workstreams that run partly in parallel. Each has a stable code (A–F) used throughout this plan and in the timeline.

ID	Workstream	Scope
A	Core Protocol (Daml)	Implement the four core templates, the *dynamic* pm-AMM pricing functions (Gaussian invariant, normal-distribution and fixed-point math, and the iterative invariant / optimal-reserve solver), and the `ResolutionPolicy`; build the Daml Script test suite covering pm-AMM invariants (price simplex, marginal price Φ, value-based liquidity, time-decay $L\sqrt{T-t}$), the lifecycle, and edge cases.
B	Resolution Infrastructure	Implement tiers T0–T3 on-ledger and the off-ledger resolver/reporter service; onboard a test source, pin its key (T1), and configure the reporter set and quorum (T2).
C	Frontend & Wallet	Build the trading frontend (market browse, buy/sell, portfolio) and integrate the Cantor8 Enterprise Wallet, including collateral transfer and reconciliation.
D	Deployment & Pilot	Continuous integration, DAR build and version management, DevNet deployment, the read-model (PQS) indexer and dashboards, and operation of the pilot markets with observability.
E	Model Marketplace	Implement the marketplace contracts, proper-scoring distribution, the reference "The Model" aggregation, and a social/agent prototype.
F	Compliance, Security & MainNet Readiness	Audit-observer reporting, a security review, key-management procedures, runbooks, and the MainNet go/no-go package.

4 Implementation Timeline

The window is sixteen weeks, grouped into four one-month phases, each closing on a review gate (G1–G4). The Gantt below shows workstream scheduling; §5 binds the gates to deliverables and acceptance criteria.



Figure 2: Sixteen-week schedule. A Core Protocol, B Resolution Infrastructure, C Frontend & Wallet, D Deployment & Pilot, E Model Marketplace, F Compliance/Security/MainNet Readiness. Gates G1–G4 close each month.

Month 1 (Weeks 1–4) — Core Protocol & Tier 0

Stand up the repository, CI, and DAR build (WS-D). Implement the Daml core package and *dynamic* pm-AMM pricing — the Gaussian invariant, normal-distribution and fixed-point approximations, and the iterative solver (WS-A) — and the Tier-0 designated-operator resolution path (WS-B). Produce the Daml Script test suite for pm-AMM invariants and the full lifecycle. *Closes at G1.*

Month 2 (Weeks 5–8) — Resolution Tiers & Frontend MVP

Implement T1 (first-party `secp256k1` verification), T2 (reporter quorum), and T3 (optimistic propose/challenge/finalize with bonds) on-ledger, plus the off-ledger resolver/reporter service (WS-B). In parallel, build the frontend MVP and Cantor8 wallet integration (WS-C). *Closes at G2.*

Month 3 (Weeks 9–12) — DevNet Pilot

Deploy the DAR to Canton DevNet under `TruthOracle::devnet` and operate three pilot markets — one T0, one T2 (2-of-3 reporters), one T3-over-T2 (WS-D). Stand up audit-observer projection, position-book reporting, and dashboards (WS-F). Track pilot KPIs. *Closes at G3.*

Month 4 (Weeks 13–16) — Marketplace & MainNet Readiness

Implement the prediction-model marketplace contracts, proper-scoring distribution, the "The Model" aggregation, and a social/agent prototype (WS-E). Conduct the security review, document resolver/source key management, and assemble the MainNet runbooks and go/no-go package (WS-F). *Closes at G4.*

5 Milestones, Deliverables, and Acceptance Criteria

Each gate is met when all of its deliverables satisfy their acceptance criteria, verified against the in-memory IDE ledger and the persistent Canton sandbox (or DevNet, where stated).

Gate G1 — Core Protocol & Tier 0 (end of Week 4)

ID	Deliverable	Acceptance criterion
D-A1	Daml core package	Compiles on Daml SDK 3.4.x; produces a versioned DAR with a stable package `name`.
D-A2	Dynamic pm-AMM & lifecycle test suite	Daml Script tests pass for the Gaussian price-simplex invariant ($P = \Phi((y-x)/L_{\text{eff}})$), time-decay $L_{\text{eff}} = L\sqrt{T-t}$, value-based liquidity, buy/sell, the direct outcome-token swap, complete-set mint/merge, the pre-trading liquidity period, and the create→trade→resolve→redeem lifecycle.
D-B1	Tier-0 resolution path	Operator can resolve a market from a designated source and a holder can redeem; void path verified.
D-D1	CI & DAR versioning	Automated build/test on every change; version bump publishes a successor DAR recognised as an in-place upgrade.

Gate G2 — Resolution Tiers & Frontend MVP (end of Week 8)

ID	Deliverable	Acceptance criterion
D-B2	T1 first-party signature	A correctly signed payload resolves the market; a tampered payload or wrong key is rejected on-ledger via `secp256k1`.
D-B3	T2 reporter quorum	M-of-N agreeing attestations resolve; disagreement or sub-quorum does not resolve and the market can be voided after grace.
D-B4	T3 optimistic flow	An unchallenged proposal finalizes after the window; a challenge escalates to T1/T2 and the losing bond is slashed to the winner.
D-B5	Resolver/reporter service	Service fetches a test source over TLS, parses, signs, and creates a valid `OutcomeAttestation`; runs one instance per reporter.
D-C1	Frontend MVP	Browse markets, execute buy/sell through the pm-AMM, and view positions on the sandbox.
D-C2	Wallet integration	Connect the Cantor8 Enterprise Wallet; collateral transfer and reconciliation complete for buy/sell.

Gate G3 — DevNet Pilot (end of Week 12)

ID	Deliverable	Acceptance criterion
D-D2	DevNet deployment & pilot markets	DAR deployed under `TruthOracle::devnet`; three markets live (T0; T2 2-of-3; T3-over-T2).
D-D3	Observability & dashboards	Trade and resolution metrics are visible; pm-AMM invariant integrity is independently checkable.
D-F1	Compliance reporting	A designated audit observer can retrieve the full position book for its market(s).
—	Pilot KPIs (tracked)	Progress toward ≥ 90 distinct trader parties, ≥ 500 `Buy` transactions, and zero failed-resolution incidents.

Gate G4 — Marketplace & MainNet Readiness (end of Week 16)

ID	Deliverable	Acceptance criterion
D-E1	Marketplace contracts & scoring	Forecasts commit only while a market is open; Brier scoring and pool distribution are tested and reproducible by an audit observer.
D-E2	"The Model" & social prototype	A reference forecast is published; a bot posts market state to a test social channel.
D-F2	Security review & key management	Review report delivered with findings triaged; resolver/source key custody and rotation documented.
D-F3	MainNet readiness package	Deployment runbook, rollback procedure, and a go/no-go checklist are delivered (execution gated on external licensing).

6 Dependencies and Assumptions

External dependencies.

- CIP-56 token semantics available on the target network for collateral denomination.
- Access to the Cantor8 Enterprise Wallet SDK and a test wallet environment.
- Canton DevNet access and allocation of the `TruthOracle::devnet` operator party.
- A test data source for resolution (e.g., an Opta sandbox/test feed) and, for T1, a signing arrangement that publishes a pinnable public key.
- Nomination of the independent reporter parties used for T2 quorum during the pilot.

Assumptions.

- Target platform is Daml SDK 3.4.x on Canton; `DA.Crypto.Text` (`sha256`, `secp256k1`) is available in the standard library.
- Daml `Decimal` (38-digit fixed precision) is sufficient for the dynamic pm-AMM normal-distribution approximations and invariant solver; no external numerical library is required.
- The team is staffed per §8 for the full window.
- DevNet and the test source feeds are reasonably available through Month 3.
- Regulatory licensing proceeds on a separate, parallel track owned by Cantor8.

7 Risk Register

Risk	Impact	Likely	Mitigation
Data source cannot sign (T1 unavailable)	Med	Med	Fall back to T2 reporter quorum, which needs nothing of the source; pursue T1 signing as an enhancement.
DevNet instability or downtime	Med	Med	Develop against the local sandbox; treat DevNet as a deployment target, not a development dependency.
Cantor8 wallet SDK access delayed	High	Low	Build C against a mock wallet adapter behind an interface; swap in the real SDK without UI rework.
Reporter parties not nominated in time	Med	Med	Run the pilot's T2 market with internal reporters first; substitute independent parties before MainNet.
Scope creep from new market types	Med	Med	Hold scope to binary markets per the proposal; log scalar/multi-outcome as a follow-on.
Dynamic pm-AMM numerical precision / solver convergence in Daml `Decimal`	Med	Low	Optimal reserves, liquidity sizing, and pool value are *closed-form* in price (x^*, y^*, V via Φ^{-1}); only the swap direction needs a guarded one-dimensional root find. Use vetted rational approximations for $\Phi/\phi/\Phi^{-1}$ with bounded error; property-test invariants and pin tolerances; fall back to a coarser static-L profile at precision limits.
Security review surfaces blocking findings	High	Low	Schedule the review early in Month 4 with buffer; triage and fix critical findings before G4.
Licensing slips (external)	High	Med	Engineering delivers a MainNet-ready package at G4; launch decoupled and gated on Cantor8's licensing track.

8 Team and Resourcing

The plan is delivered by a two-person senior team across the window:

- **Lead Engineer** — owns the Daml core protocol and resolution (WS-A, WS-B), the off-ledger resolver/reporter service, continuous integration and DevNet deployment (WS-D), and the security review and compliance reporting (WS-F). Accountable for the gate deliv-

erables.

- **Full-Stack Engineer** — owns the trading frontend and Cantor8 wallet integration (WS-C) and the prediction-model marketplace and social/agent prototype (WS-E), and builds the observability dashboards within WS-D.

The two engineers share WS-D and WS-F operationally; all backend and service work is absorbed into these two roles. Should Cantor8 wish, an independent security reviewer may be commissioned for WS-F, but the plan does not assume one.

Cantor8 provides the wallet SDK, DevNet access and party allocation, data-source and reporter introductions, and the business/compliance counterpart for gate reviews.

9 Review Cadence and Governance

- **Bi-weekly demo** — a working build is demonstrated every two weeks against the sandbox or DevNet.
- **Monthly gate review (G1–G4)** — at each gate, deliverables are checked against their acceptance criteria and signed off jointly before the next phase begins. A gate not met is re-scoped at the review rather than silently carried forward.
- **Change control** — scope changes are recorded against this annex with their schedule impact, so the agreement and the plan stay consistent.

10 Exclusions (Out of Scope)

The following are explicitly outside this engagement and are noted so the agreement is unambiguous:

- Regulatory licensing in any jurisdiction, and the production MainNet launch, which is gated on that licensing.
- Build-out of a KYC/AML provider; the plan integrates with Cantor8's existing arrangements rather than building one.
- Fiat on/off ramps and collateral-token issuance.
- Native mobile applications (the frontend is web).
- Formal verification or a full external audit beyond the scoped Month-4 security review.
- New market structures beyond binary markets (e.g., scalar or multi-outcome), which are a follow-on per the proposal.

Document control

Plan v1.2 (22 July 2026). Four-month technical implementation plan implementing TruthOracle proposal v0.7; for attachment to the delivery agreement. Tracks the v0.7 specification: the pm-AMM workstream (A) and acceptance criteria now cover the reference-implementation execution model — complete-set mint/merge, the direct outcome-token swap, cached-price / virtual-reserve execution, and the pre-trading liquidity period for dynamic pools. (v1.1, 27 June 2026: revised for the dynamic pm-AMM, the read-model (PQS) indexer, and the operator-console / key-management control plane; v1.0, 10 June 2026: initial plan.)

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